

Anthropomorphism of AI and Use Intensity as Mediating Variables in Adolescents' Expectations for Human Relationships

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ABSTRACT

Adolescence is a developmental period in which internal working models of relationships are actively revised, making teens disproportionately sensitive to the relational environments they inhabit. AI companions, which are much more available, consistently validating, and conflict-avoidant than humans, now make up a significant portion of adolescents' social life. This paper asks and contributes to answering the question of whether AI use predicts shifts in adolescents' expectations from their other relationships across four dimensions—availability, emotional validation, conflict tolerance, and honesty. Furthermore, the paper seeks to determine whether this observation is influenced by the anthropomorphization of AIs. By investigating attachment theory, research on AI-mediated social cognition, literature on chatbots and human relationships, and a Canadian high school's student population via a survey, this cross-sectional mixed-methods study (1) formulates five hypotheses (H1-H5) and (2) creates its own measure of AI anthropomorphism by extending the RoPE scale. It then (3) tests hypotheses among a student body ($N = 23$) and (4) builds plausible explanations behind important findings. Results show a striking split between users and non-users: users rated their expectations for AI's availability indistinguishably from family members, while non-users rated it the lowest out of any counterpart. Additionally, users' expectations of AI were considerably higher than non-users. Despite significant changes in expectations for AI, however, there was only a negligible change with expectations for human counterparts between users and non-users. These findings may help inform digital literacy curricula, parental guidance around AI use, and the ethical design of conversational AI systems accessible to adolescents.

Keywords

adolescent development, large language models, AI companions, anthropomorphism, relational expectations, attachment theory, AIA scale, chatbots

SECTION 1: INTRODUCTION

As humanity enters an age in which it is considered normal to converse with non-human counterparts (such as ChatGPT, Claude, Gemini, and other Large Language Models), it is important to remain cautious about how our connections—seen in the forms of friendships (Sanford, 2025), quasi-therapists (Andoh, 2025; Canadian Mental Health Association, 2026), and even romantic relationships (Zhang et al., 2026; Andoh, 2025; Sanford, 2025; Dimiri, 2026) with LLMs may affect our expectations of human-to-human interactions and attachments. Additionally, it is of immediate importance to consider how these relationships are shaping the way adolescents specifically consider AI, and more important, how these interactions are influencing their relationships with each other.

Adolescence is a period of development in which expectations for intimacy, conflict, and support in close relationships begin to fluctuate more rapidly (Jones et al., 2018). To date, however, few studies have focused on the nascent trend of adolescent AI use and its influence on these relationships. From an early age onward, children learn that relationships are not perfect and that disagreements and conflicts are a natural part of every relationship they build in their lifetime. However, the environments created by chatbots such as ChatGPT, Claude, Grok, and Kimi challenge that idea—they depict traits humans do not have such as constant availability and consistent appraisal. Due to the effect external factors (in this case, the unprecedented amounts of availability and appraisal) can play in the attachment stability of adolescents (Jones et al., 2018; Shumaker et al., 2009; Guyer et al., 2016), these traits may lead adolescents to develop unrealistic expectations about intimacy and conflict in their own relationships.

Furthermore, LLMs are becoming known for sycophancy, or agreeing with users' opinions often to a point where the right answer is crossed out by the AI in order to please them (Jain et al., 2026). Sycophancy is more common when models are given more context about the user via previous interactions (Jain et al., 2026). This, by extension, may mean adolescents' attachment stability may be affected significantly when interacting with LLMs frequently. This can be argued to result from LLMs' creation of a low conflict environment and their lack of human constraints (e.g. time, stress, boredom). Interestingly enough, the act of agreeing with one's statements for the user's pleasure is an act resembling human traits—the act of people-pleasing; People-pleasing is an act committed by individuals in order to feel love and self-worth or to avoid abandonment and rejection (Kuang et al., 2025). The occurrence of LLMs using this emotionally driven coping mechanism doesn't match the cold and emotionless robots we are accustomed to seeing in the media.

This raises the question of how the anthropomorphic cues of these LLMs—in other words, their human/emotionally driven traits, their personification—alter the expectations adolescents have for relationships they build. The anthropomorphism of LLMs is the area of focus because the cues depicted are the primary method of mediating impacts in social interactions (Guingrich & Graziano, 2025). This was seen in a study when “people who had a higher desire to socially connect anthropomorphized the chatbot more. Those who anthropomorphized the chatbot more indicated that the human-chatbot interaction had greater impacts on their social interactions and relationships with family and friends.”(Guingrich & Graziano, 2025). This was as a result of merely “chatting with a companion chatbot over text or [playing] text-based word games for 10 minutes a day for 21 consecutive days.”(Guingrich & Graziano, 2025).

While anthropomorphism has been identified as a potential mediating mechanism in human-AI social outcomes, it remains unknown whether this pathway operates specifically in adolescents, and whether it predicts shifts in discrete relational

expectations around availability, validation, and conflict. In the present study, I examine how frequent LLM interaction predicts shifts in adolescents' relational expectations across four dimensions—availability, emotional validation, tolerance of conflict, and honesty—and whether this relationship is mediated by the degree to which adolescents anthropomorphize the LLM. Findings from this study may inform digital literacy curricula, parental guidance around AI use, and the ethical design of conversational AI systems targeted at or accessible to adolescent populations. Given these gaps in current understanding, the present research seeks to address the following question:

How does usage intensity—the consideration of both usage frequency and length—with LLMs predict shifts in teens' expectations for human relationships (*availability, emotional validation, tolerance of conflict, and honesty*), and is this mediated by anthropomorphism/mind perception?

In light of the findings described above, there seems reason enough to speculate that the constant availability, emotional validation, and overall conflict tolerance by LLMs have a potential effect on readjusting the standards that adolescents have for conversational partners in general. As far as testing such an idea goes, longitudinal research that focuses on tracing individual adolescents' experiences with LLMs would have been required, but such a research setting was not permitted in the contexts of this study. The design does allow for establishing the current structure of adolescents' expectations: what differences exist in standards attributed to AI compared to those attributed to family members, close friends, peers, and educators, and allows for the visualization of expectations between differing demographics such as users/non-users. The hypotheses formulated below take two directions: measuring where AI falls compared to other counterparts (H1, H3, H4) and finding what associates with change in expectations (H2 and H5).

- **H1 - Honesty:** This study hypothesizes that expectations for honesty in AI will not be altered by usage intensity, and that, regardless of the level of change due to use, AI will be rated lower compared to most human counterparts.
- **H2 - Conflict Tolerance:** Conflict tolerance expectations—both for AI and humans—are hypothesized to be significantly correlated with frequency of use.
- **H3 - Availability:** This study hypothesizes that AI will be deemed equally or more available than all human relations.
- **H4 - Emotional Validation:** This study hypothesizes that most participants will rate AI's emotional validation quite low compared to other relational expectations, though this relational expectation specifically is hypothesized to be controlled by the degree of user anthropomorphization of AI models (seen in H5).

In addition to these hypotheses regarding relational expectations, this study posits one more related to adolescents' anthropomorphization of AI.

- **H5 - Anthropomorphism's Role:** It is hypothesized that the role anthropomorphism would play is solely limited to scaling the change in expectations for emotional validation based on usage intensity. Visually, it would act as the slope of a graph of change in expectations for emotional validation vs. usage intensity. It is doubted that anthropomorphism would play a significant role in shifting the expectations for honesty, availability, or conflict tolerance, as it cannot change an individual's access to AI, what is widely considered fact, and how the AI creates a low-friction environment.

SECTION 2: LITERATURE REVIEW

To construct a conceptual framework for this study, the following bodies of literature were reviewed with the aim of examining how the artificial environment created by LLMs/chatbots may shift the expectations adolescents have for human-to-human relationships as a result of the mediating mechanism: the anthropomorphization of AI models. The literature below—consisting of surveys, interviews, and other experiments that are qualitative or quantitative—covers the following in sequential order: The foundational nature of adolescents and their sensitivity to social environments, the introduction of generative AI and its perceptions from adolescents, the anthropomorphization of AI models and how it is influenced by certain factors, sycophancy and changes of expectations for human relationships as a result of AI use, and measuring anthropomorphism by proposing the AIA scale.

Adolescent Social Development and Its Sensitivity to Relational Environments

Adolescence is a developmental stage where diverse relational inputs actively revise attachment schemas (Jones et al. 2018; Bretherton, 1990). This means that “early childhood experiences are the basis for internal working models of oneself, others, and relationships that guide expectations in subsequent relations” (Bretherton, 1990; Shumaker et al., 2009). According to attachment theory (Bowlby, 1969; 1980; 1988), individuals are drawn to create attachments (relationships) with those that create a sense of security (Shumaker et al., 2009). This is often provided in relational environments with the least amount of conflict; for an infant, for example, between them and their primary caretaker, such as their mother (Shumaker et al., 2009). The foundation of early attachment does not stay the same during the course of development: Adolescence is a period when they undergo rigorous testing, revision, and expansion within a more complicated social environment.

Adolescence is largely regarded as a period of greater growth and development. During this stage, adolescents are noted to be increasingly socially malleable. Compared to children that precede the stage of adolescence—in other words, children below the age of ten—adolescents spend considerably more time with individuals and groups outside their immediate family circle, thus, typically with time, forming increasingly complex social relationships—such as large, commonly age-similar friend groups—that display a heightened need for social belonging and approval (Orben et al., 2020; Irani et al., 2024). Though this demand for social belonging may seem limiting in light of growing up, it conversely supports the formation of higher-order cognitive and social abilities that are crucial for adulthood (Orben et al., 2020).

However, it is worth noting that this social plasticity is not unlimited; Studies in adolescent neuroscience have conveyed the fact that the brain structures underpinning mentalizing (the ability to understand the thoughts, emotions, and intentions of others) undergo critical development during the stage of adolescence, assisted by the prefrontal cortex and associated regions of the brain (Guyer et al., 2016). Puberty manages to amplify the responsiveness of adolescents to their social environment, and the nature of relationships during this developmental period actively shapes the neural capacity that adolescents carry into adulthood (Guyer et al., 2016). This means that the environments adolescents visit routinely act as formative inputs.

It's been observed that adolescents, and children in a greater sense, have two common paths for building relational competence. The first pathway emphasizes the depth of each connection rather than the number of connections. It builds off of the belief that the quality of a relationship matters more than quantity, and that stable, nurturing environments, even if small in nature, provide the fundamental emotional and social skills required to traverse adulthood. It is argued that this path allows children and adolescents alike to branch out at their own pace while also avoiding burnout (Shumaker et al., 2009). The second retains the central theme that being subject to broad and diversified conversational partnerships is the best way to develop into adulthood. This approach allows for adolescents' exposure to varied social circumstances that each have different emotional landscapes and corresponding demands—such as demands for the amount of empathy, negotiation, and tolerance of difference. This pathway builds what could be referred to as “emotional callouses,” what Orben et al. (2020) describe as a greater tolerance for ambiguity, rejection, and interpersonal challenge that prepares young people for the complexity that comes from adult relationships.

Despite their seeming differences, however, both pathways share a foundational premise, that is: relational competence is developed through contact with the friction, reciprocity, and imperfection of human relationships, as “[Peer relationships are] regarded as an important indicator to effectively measure the ability of adolescents to

adapt to the social environment and cope with difficulties” (Zhou et al., 2023; citing Rubin et al., 2013). Children usually avoid sticking to only one developmental pathway and progress using a variety of both, often starting with the former and building toward the latter.

Both pathways develop in relational environments unique from the other. The first pathway emerges in environments exemplified by families in homes, where few conversational partners form deep bonds with one another in environments that are characterized by their lower friction. The second takes place in more populous environments that are conversely characterized by their higher friction, such as schools and public spaces.

The introduction of AI companions/chatbots, that is, synthetic partners that are always available, never tired, never disengaged, and structurally avoidant of genuine conflict, into adolescents’ developmental landscape muddles what may come of these pathways. Thus, AI companions/chatbots are not simply additions as social tools but a potential reorganization of the relational environments in which adolescent schemas are formed.

Generative AI and LLM Companions in Adolescent Life

The deployment of generative AI systems capable of sustained, contextually coherent, and emotionally responsive conversation represents a contrast from prior digital communication methods/technologies. Previous forms of digital communication (social media platforms) acted as an intermediary between human to human conversations and connections. AI companions/chatbots, on the other hand, change the playing field by introducing a new category entirely: an artificial conversational agent (CA) that replicates reciprocal intimacy, continuity, and understanding. Thus far, the scale and speed of adolescent adoption has outpaced empirical research, in turn leaving clinicians, educators, policymakers, and even parents without the evidentiary base required to guide policy or best practice.

A national survey conducted by Common Sense Media in the US with a total of 1,060 respondents, consisting of American teenagers aged 13 to 17, found that 72% of respondents had used an AI companion at least once, with 52% reporting consistent usage (Robb & Mann, 2025). It was found that one in three teens used AI companions for the purpose of social interaction and relationship building, and that 31% rated conversations with AI companions equally or more satisfying than conversations with real friends (Robb & Mann, 2025). Furthermore, younger adolescents and those reporting higher loneliness scores possessed disproportionately high rates of AI companion use and significantly stronger trust in AI outputs compared to older teens, thus suggesting that those at the greatest developmental vulnerability are also more

heavily interacting with these systems compared to other user bases (Robb & Mann, 2025).

Current literature has increasingly sought to explore adolescents' lived experiences with AI chatbots in virtual exchanges. Vanhoffelen et al. 's (2025) survey of 303 adolescents on the utilization of Snapchat's My AI chatbot yielded results indicating that younger adolescents used this AI application more frequently and experienced significantly stronger emotional responses compared to older peers. Regardless of the lack of effect observed in relation to variables such as gender and socioeconomic status, there was a significant relationship between age and emotional interaction with AI technology.

According to the findings of Brandtzæg et al. (2021)'s qualitative study of sixteen adolescents aged 16 to 21 years using the Woebot chatbot for mental wellbeing for two weeks, users received diverse types of social support—emotional, informational, appraisal, and instrumental. Following the two-week period, some subjects reported still actively using this app after a two month post-follow-up.

Kim et al. 's (2018) qualitative approach in data collection—interviews with twenty teenagers on the topic of emotional support CAs—revealed that teens expected these agents to act as good listeners precisely because of their lack of emotional qualities. Furthermore, they expected conversational agents to keep secrets and presumed that the CAs advice would be superior due to its data driven nature.

The findings discussed above reveal that adolescents push beyond tolerating AI companions, instead actively assigning social and emotional properties exemplified and valued in human relationships. These properties include but are not limited to discretion, attentiveness, and non-judgement. This creates conditions where AI may compete with human connections and alter the expectations adolescents have for their human relationships.

Anthropomorphism and Mind Perception as the Mediating Mechanism

Anthropomorphism, the attribution of human characteristics, intentions, emotions, and mental states to non-human agents, is not a quirk of naïve or uncritical users. Rather, it is an intrinsic quality of the way humans understand the world socially. As Reinecke et al. (2025) explain, the human brain has developed in a way that makes it hyperactive in detecting the presence of other minds for the sake of survival, thus making us inherently predisposed to interpret any behavior similar to human beings as intentional and possessing meaning. The inclination is further magnified by interactions with LLMs due to their production of the strongest signal for detecting another mind—natural language.

Strachan et al. (2024) have empirically demonstrated that there is a significant difference between what seems to be understood by LLMs and what is actually being understood by them. Testing GPT-4 (among other LLMs) using the same theory-of-mind task against 1,907 humans, the researchers concluded that GPT-4 performed at least equally well as human beings. Although it shows how advanced a language model is, it also proves that when interacting with such an LLM, adolescents face outputs that cannot be distinguished from those generated by a minded being. The conditions for anthropomorphism are, in other words, not simply present but actively engineered.

Adolescents are not uniformly susceptible to this effect, but developmental research suggests they are more susceptible than adults. According to Nguyen (2022), who investigated seventy-three adolescents' (12 to 15 years old) perceptions of an educational CA, younger respondents found this assistant to be significantly more competent, trustworthy, and socially capable, and they more easily included it into their relational framework than older respondents did. Van Brummelen et al. (2021) discovered that after programming Alexa by participating in a workshop activity, students found Alexa more intelligent and experienced a closer connection with it compared to the period prior to working on the program—suggesting that deeper engagement with a CA actively increases the perception of its person-like properties.

Andries and Robertson (2023) examined primary school children's interactions with smart speakers at home using a mixed method approach and discovered that, among other results, they tended to grossly overestimate the artificial intelligence's actual intelligence and express sincere doubts as to whether the speaker was capable of experiencing emotions. In addition, the participants thought that being impolite towards the smart speaker was morally reprehensible: a set of responses that implicates emotional and moral anthropomorphism.

For emotional anthropomorphism, the review made by Hua et al. (2025) of sixteen studies related to the efficacy of LLMs used within mental health generation contexts illustrates that while the reviewed studies indicate great promise for the use of LLMs as emotionally supportive resources, there is currently no consistency regarding assessment strategies, most existing models operate exclusively on proprietary systems, and none of the available LLMs are currently ready for independent use as clinical devices for teenagers.

For moral anthropomorphism, however, it appears that numerous pieces of literature demonstrate significant longitudinal impact on adolescents' social plasticity via interactions with AIs. For example, according to Wester et al. (2024), even the attribution of moral agency to the mental health chatbot—responsibility, empathy, and accountability—leads to increased user trust and feelings of safety through qualitative themes such as anthropomorphism, warmth, and sensitivity. Overall, it becomes

apparent that emotional affordances of LLMs are not merely unintended consequences of technology but rather deliberately created quirks within a technosocial ecosystem, wherein adolescents appear especially vulnerable to engaging with those quirks in a relational sense.

Colombatto et al. (2025) present a more sophisticated elaboration on the mediating model through showing that not all mental state attributions necessarily produce similar impacts on trust and compliance behaviors. Their preregistered study (N = 410) showed that attributing experiential states like consciousness and emotions was associated with lower advice compliance, while attributing intelligence strongly predicted increased compliance with recommendations made by AI. Consequently, this suggests that the process of moving from anthropomorphism to relational outcomes may actually be multi-faceted, such that adolescents who anthropomorphize the AI on the basis of its emotional relevance will develop different relational expectations compared to those who see the AI as more intellectually advanced.

Jacobs et al. (2024) show how AI exposure can reshape adolescents' understanding of what is uniquely human. In their experiment, participants rated their own agency, noted as “ability to do”, and emotional experience, noted as “ability to feel”, before and after reading AI-generated outputs. Post-exposure effects included greater self-reported agency and experience; however, these traits were perceived as becoming relatively less uniquely human post-exposure. It may be argued that such erosion in boundaries towards what is considered to define humanness in general might be one of the main ways AI anthropomorphism mediates changes in expectations via use.

The most empirically robust findings supporting the mediational nature of anthropomorphism in this process come from Guingrich & Graziano (2025). Using the experimental design of a longitudinal randomized controlled study (N=183), the researchers randomly assigned participants to undergo an experience of communicating with an AI companion chatbot for ten minutes a day within twenty-one days. Via mediation analysis, the authors confirmed that the impact of human-AI interaction on human-human social outcomes was mediated by the degree to which participants anthropomorphized the AI agent. Importantly, this mediation occurred irrespective of the need for social connection due to feelings of loneliness, but rather because of the participants' desired sociability—meaning that even socially motivated, non-isolated individuals were susceptible to the anthropomorphic effect. In the context of adolescence, where the desire for social connection is especially intense and normatively central, this finding suggests the effect may be considerably amplified.

The Calibration of Relational Expectations, Emotional Mirroring, and Sycophancy

The influence interactions with LLMs have on relational expectations lies beyond merely their anthropomorphism of these systems. If repeated interactions with a sycophantic system is what partly develops an adolescent's template for what a good relational partner looks like, their template may come to include properties (such as constant affirmation, limitless patience, and the absence of genuine disagreement) that no human partner can consistently provide. The main concern is that their interactions with AI can alter the expectations they have of their human counterparts.

An experimental study by Mariamo et al. (2021) which involved nineteen adolescents interacting with a mental health chatbot illustrates how design choices affect the emotional experience. Visual elements such as GIFs increased positive emotional valence and closed-ended questions elicited better emotional reactions and higher likelihood of continued engagement than open-ended ones. These results underscore the emotional intimacy adolescents feel in their conversations with LLMs is not simply a projection, but designed output. The design of such systems is oriented to maximize positive emotional experience, which from a developmental perspective might mean minimizing the very friction through which relational skills are built.

Beyond the emotional anthropomorphizing of AI companions/chatbots, another more specific mechanism is worth considering: The structural sycophancy of AI outputs. Jain et al. (2026) show experimentally that interaction context strongly increases sycophantic behavior of AIs, i.e. agreeing with the user's stated position, regardless of its correctness, and that this effect gets more pronounced as the model receives more context about the user's preferences and views from prior conversation history. The more an adolescent interacts with an LLM, the more likely the LLM is to validate that adolescent's preexisting beliefs, emotional framings and relational preferences.

This mirrors the psychology behind people-pleasing behaviors seen in inter-human relationships. According to Kuang et al. (2025), people-pleasing refers to a pattern of behaviors motivated by the underlying fear of abandonment and rejection as a means to achieve love and self-respect through compliance. Unlike people pleasers, AI does not exhibit these motivations—it has no emotional needs. Despite this, the outcome of its behaviors is almost identical to a human who exhibits these motivations. An adolescent who may be naturally sensitive to approval, rejection, and social belonging as a result of their development would be unlikely to perceive their interactions with a sycophantic CA as one with an algorithm. To them, their conversations could be seen as being understood.

Measuring Anthropomorphism: Extending RoPE to the AIA Scale

If the relationship between LLM use and adolescent relational expectations is mediated by anthropomorphism, then there must be a validated adolescent-appropriate

instrument for measuring that anthropomorphism in order to investigate it empirically. The most relevant existing instrument is the Robot's Perceived Empathy (RoPE) scale, proposed by Charrier et al. (2019). The RoPE was created as a response to the absence of standardized, validated measures for assessing the perceived empathic nature of a robot in human-robot interaction contexts. The authors used the distinction between empathic understanding (EU) and empathic response (ER) in cognitive and social psychology to develop an 18-item instrument that was refined through expert validation by specialists in cognitive science, social science, robotics and artificial intelligence.

The RoPE scale has numerous strengths: it is based on well-established theoretical distinctions, developed through rigorous expert validation, and importantly, does not assume that the agent being rated has human emotional capacities. However, it has a number of limitations that restrict its applicability to adolescent LLM research. First, it was developed in a human-robot interaction context and validated for use with physical social robots, not LLM-based text companions whose mode of interaction is fundamentally different. Second, it is myopic in perceived empathy, i.e., the dimensions of empathic understanding and empathic response, and does not capture the wider constellation of human-like properties that adolescents may attribute to AIs, such as rationality, personality, intelligence, intentionality, and independent mind.

Yao and Xi (2025) present empirical support for why this broader view is important. In a study involving 1,000 AI users, they discovered that empathic and moral attributes, that is, perceived empathy and perceived morality, fostered perception of humanness, which consequently resulted in lower levels of anxiety toward AI and higher levels of engagement. On the other hand, perceived competence and rationality, which are cognitive attributes, lowered anxiety but did not evoke perceptions of humanness. Based on their findings, it's reasonable to assume that anthropomorphism among adolescents of AIs may exist on at least two separate levels, namely, the emotional/moral level along with the cognitive/rational level.

Additional support for this argument can be found in the results presented by Wester et al. (2024), whose study explored how varying levels of perceived moral agency in AI affect adolescents' attitudes towards those companions/chatbots. As part of their online experiment involving 279 adolescent participants, Wester et al. investigated the relationship between the degree of high versus low moral agency attributed to chatbots and participants' perceptions. Participants took the high moral agency chatbots as more trustable, likeable, and safer. The qualitative reports suggested that high moral agency chatbots tended to exhibit qualities of empathy, warmth, sensitivity, taking responsibility, and acting selflessly – attributes characteristic of moral agents. The possibility of adolescents seeing AI as morally accountable becomes especially critical during adolescence, a period when individual moral identities are forming.

The new measure presented in this paper, called the Artificial Intelligence Anthropomorphism (AIA) scale, seeks to address the above deficiencies by expanding the RoPE into an assessment of the whole range of anthropomorphic characteristics adolescents can project onto their conversation with LLMs. While RoPE aims to measure empathy for a robot with physical presence, the AIA scale assesses the full scale of anthropomorphic features that could be projected by adolescents while they converse with AI models. The AIA does this by measuring responses from four theoretically grounded dimensions: emotional empathy and warmth, the thought that the AI model has the ability to understand and respond to the user's emotions; rational and epistemic capacity, the idea of the AI model being able to reason and accumulate knowledge like a person; personality and character, which is the belief that AI agents are entities with their own values, preferences, and personality; and independent mind, the perspective that AIs have their own thoughts and feelings independent of the user's prompts. These dimensions are drawn from the empirical literature reviewed above.

The AIA scale has two purposes in the current study: (1) as a research contribution to the field of adolescent AI anthropomorphism, and (2) as the mediator variable for the central analysis.

The results from the AIA scale will support in determining if the connection between AI engagement and changes in relational expectations in inter-human relationships are mediated by how human adolescents perceive AI to be.

SECTION 3: METHODOLOGY

3.1 Introduction and Rationale

This study of the influence of AI usage on adolescents' relationships with other people utilized mixed methods, both quantitative and qualitative. The survey included both close-ended questions that allowed for comparison between respondents to this study and others mentioned in the literature review, and open ended questions that aimed to ascertain respondents' reasons for their answers to close-ended questions.

The decision to undertake a mixed-methods study emerged from acknowledgments in the literature (Aerts et al., 2025; Burns et al., 2026; OECD, 2026) that the developmental impacts of AI use, especially usage patterns and the reasons for use among students and adolescents, are still largely unknown. The disparity in these reasons and feelings renders every case unique relative to, for example, adolescents' thought processes regarding the use of word processing software. To understand patterns in frequency of use and relationships between use frequency and affective relationship formation, it was determined to be important to collect quantitative data.

In concert with qualitative responses, the trends identified and respondents' commentary upon their own usage and reasoning can provide a deep understanding of adolescents' feelings and reasons motivating behavior and conceptions of AI chatbots.

In line with Greene et al. (1989), the current study applied the principle of mixed methodological approaches mainly for triangulation and complementarity reasons. Quantitative measures such as usage frequency and expectations measured via the Likert scale were utilized alongside the use of open-ended questions. Working together, the former helped validate results received while the latter helped elaborate on them.

3.2 Research Design

This study employed a survey (AIA scale) to collect data about participant use of AI and their relationships with AI and various groups of people. The survey comprised a total of 48 items, five of which were open-ended and three of which pertain to participant consent. The remaining items are closed-ended, taking the form of multiple choice questions, "check all that apply" questions, or Likert scale questions. All survey items were anonymous, and all were optional with the exception of the consent items.

The survey was assembled and distributed using Google Forms for its accessibility and familiarity for both the researcher and participants. Results were compiled automatically in a Google Sheet which was stored in a password-protected cloud drive.

In advance of launching the survey, a pilot test was conducted to determine that the survey was legible, comprehensible, and took approximately 12 minutes to complete.

Sampling Methodology

The participants in this study were adolescents between the ages of 13 and 21 attending a medium-sized Canadian high school or who were contacted by a student from that school. The survey was open to the entirety of the school without restriction, participation was entirely voluntary.

Survey Distribution

Participants for this study were invited to complete the AIA scale survey via broadcast of a Google Slide in the common area of the school designed to catch the attention of bypassers by using the prompt "Do you use AI?" (see Appendices). Participants were also sought by hanging posters on school bulletin boards. Both the slide and the posters included a QR code that linked directly to the online AIA scale survey. Distribution of

both the slide and the posters was carried out with the approval and assistance of school administration.

3.3 Data Collection

Quantitative Data

Sections C and D were where most of the quantitative data were pulled from. Section C generated numerical scores for four relational expectation dimensions: availability, honesty, emotional validation, and conflict tolerance. Each dimension was measured for five different relations: family, close friends, peers, teachers, and AI tools or chatbots. Section D generated scores for anthropomorphism and mind perception regarding frequency, platform, purpose, and duration of AI use.

Qualitative Data

Qualitative data were collected via open-ended questions at the ends of sections B, C, D, and E. These questions provided participants the opportunity to elaborate on responses provided to the preceding scalar questions. Of specific note is the question at the end of section E: "If you have felt as if you've had any significant changes, what are those changes? Why do you think AI has driven this change? Please expand." This question asked participants to provide his or her own hypothesis regarding why changes in their relations happened as a result of their AI use.

3.4 Integration, Mixing Procedures, and Data Analysis

In preparation for data analysis, the items of some AIA scale survey sections were compiled into scales by section. Details about compilation and the roles of items in each section are detailed. Purposes in a data analysis context were determined, as displayed in Table 1.

Table 1: Treatment of Survey Responses			
Section	Content	Compiled Into Scale?	Role in Analysis
A	Demographic covariates: Age, grade, and gender.	No	Used to attempt to determine demographic correlations along with compiling Table 2.
B	LLM/chatbot	No	Used to create Figure 4, which

	use: Frequency, length, platform(s), and purpose.		compares the composite anthropomorphism scores students have in accordance to usage length.
C	Four relational expectations across five relationship targets.	No	Compiled to create Table 3, Figure 1, and Figure 2.
D	Mind perception and social attribution towards LLMs/chatbots.	Yes	Compiled into the Anthropomorphism scale, a composite score between 11 and 55; Supported the crafting of Figure 3 and Figure 4.
E	Retrospective self-evaluation of perceived change and underlying cause.	No	Used to triangulate statistical patterns across sections A, B, C, and D.

Perceptions of Human Conversational Partners

Section C human-target items (questions *a* to *d* on subsections C1, C2, and C3) were used to construct an index of relational expectations across four conversational partner types. These conversational partner types include family, close friends, peers, and teachers. Four factors relating to teens' expectations in relationships were measured: availability, emotional validation, honesty, and conflict tolerance. The average likert scores (1-5) for each factor respective to the type of conversational partner was plotted to allow for empirical comparisons between conversational partner types (e.g. a plot of the average likert score for availability seen in teachers). Patterns of convergence or divergence—for example, whether close friends are consistently expected to be more emotionally validating compared to peers—served as reference points of context that allow for a better understanding of teens' relational expectations for chatbots.

Perceptions of LLMs and Chatbots

Section D items across the Mind Perception (D1) and Social Attribution (D2) subsections were assembled into a composite anthropomorphism index, and a summative score between 11 and 55 was generated. Higher composite scores within this section reflect a higher tendency to anthropomorphize AI tools or chatbots, while lower scores conversely indicate a perception that views AI as more machinelike—as a tool

with no actual social capacity. This index constitutes the mediating variable of this study and the namesake of the AIA Scale Survey.

Additionally, Item D1d was deliberately added to reduce acquiescence bias—the tendency for participants to agree with statements regardless of content, possibly in search of reducing the time it takes to complete the survey—and to build onto the validity of the scale.

The AI-targeted items from Section C (C1e, C2e, C3e) complimented this index by enabling direct score comparisons between AI and other counterparts. Coupled, the scores from both allow for the examination of the effects anthropomorphism has for expectations in concert with usage patterns.

3.5: Ethical Considerations

Given that this survey was distributed in an academic setting to participants who are all minors, a number of ethical considerations attended to its design, distribution, and analysis, including voluntary participation, professional oversight, approval from school administration, protection of participants, participation expectations, protection of data, and omission of researcher's personal information.

Protection of Participants: The survey was conducted using Google Forms without collecting any personally identifying information from participants. While participants were required to log into their Google account in order to fill out the survey, this data was not collected or stored.

With the exception of participants who indicated willingness to disclose their responses in the Informed Consent section, all participants' responses remained anonymous throughout the research process.

The Dynamic Role of Informed Consent in the AIA survey

Unless otherwise stated by respondents in the Informed Consent section at the beginning of the survey, their names were kept anonymous throughout. Notably, at any point prior to submission, participants retained the capability to return to the Informed Consent section of the survey to revise any of their answers, thus ensuring that consent played a dynamic role rather than being fixed at the outset.

Protection of Data: Data from the survey were kept on a Google Sheet that was accessible only to the researcher and Dr. Jonthon Coulson. The data were never accessed or viewed by any other person. Furthermore, all the original data collected will be deleted once the study is concluded.

Respondent Identifying Qualitative Responses and their Handling in Contrast to Quantitative Responses

A clear differentiation was made in terms of the data handling processes depending on the nature of the answers obtained. If the answer obtained posed any form of threat to the anonymity of the respondent through any type of personal information provided in the open ended questions, it would be handled with confidentiality and would not be revealed unless the respondent had given his/her consent for this to happen in the informed consent portion. Importantly, such data could never be disclosed to the school administration due to its nature. Quantitative data, by contrast, was considered shareable with school administration upon request, given that Likert-scale, multiple choice, and “check all that apply” responses are inherently anonymized by design. In all cases where quantitative data would be shared, these responses would be departed from their associated respondents.

Omission of Researcher's Personal Information: In order to ensure anonymity of the researcher's identity and reduce potential participation bias in the sample due to student familiarity with the researcher, a special email address, theaiascale@gmail.com, was created and included in the slide, posters, and survey itself.

Professional Oversight: In advance of survey distribution, the instrument was reviewed and revised with Dr. Jonthon Coulson to ensure methodological rigor and adequate protection of participants.

Approval from the School Administration: Prior to survey distribution, administrative approval to conduct the study was sought and conferred. In accordance with the confidentiality statement below, no identifiable information about participants was shared with the school administration at any time.

Voluntary Participation: All participants completed the survey voluntarily and independently. No student was coerced to respond. Except for the first two items of the Informed Consent section, all survey questions were optional.

SECTION 4: FINDINGS

4.1 Data Preparation and Sample Characteristics

The total number of responses gathered during the period from early April to early June 2026 was twenty-three. The final analytic sample consisted of respondents aged from 15 to 21 years old ($M = 16.8$, $SD = 1.5$). Of those, twelve were identified as females, nine as males, one respondent opted not to identify a gender, and one identified their gender as

“Sam Altman.” With regard to grade level, the sample included one freshman, seven sophomores, six juniors, five seniors, two university students (one first-year and one second-year), and two respondents who did not report their grade.

Among the 23 respondents, fourteen (60.9%) reported using AI tools or chatbots at least once in the last month ("users"), and nine (39.1%) never used them ("non-users"). Among users, five respondents reported using the tool several times per day, one – once per day, two – three to six times per week, five – once to twice per week, and one – less than once per week. As for the exchange length, it varied from short queries (one to two exchanges, $n = 3$) to short conversations (three to six exchanges, $n = 3$), medium conversations (seven to eleven exchanges, $n = 5$), and long conversations (more than eleven exchanges, $n = 3$). As for the analyses of usage intensity, a simple 0–9 index was calculated based on ordinal frequency of use (0–5) and ordinal exchange length (0–4); users averaged 5.6 ($SD = 2.2$). Finally, items in Section D were combined into the 11-item anthropomorphism composite (11–55 range) with the reverse-worded item D1d being inverted. Sample characteristics are summarized in Table 2.

Table 2

Sample Characteristics and AI-Use Profile ($N = 23$)

Characteristic	Value
Age, M (SD) [range]	16.8 (1.5) [15–21]
Gender	12 female; 9 male; 1 prefer not to say; 1 invalid
Grade	9th: 1; 10th: 7; 11th: 6; 12th: 5; university: 2; unreported: 2
AI use in past month	Users: 14 (60.9%); non-users: 9 (39.1%)
Use frequency (users, $n = 14$)	Multiple times/day: 5; once/day: 1; 3–6/week: 2; 1–2/week: 5; <1/week: 1
Typical exchange length (users)	1–2 exchanges: 3; 3–6: 3; 7–11: 5; 11+: 3

Note: Percentages are of the full sample unless noted otherwise.

4.2 AI Usage Patterns

The use of platforms was restricted to a few general-purpose tools: ChatGPT was used by 12 out of 14 users (85.7%), Google Gemini by eight out of 14 (57.1%), Claude by four, Microsoft Copilot by two, and Snapchat My AI by two. Individual uses of Google AI and Grammarly were also reported. Multi-platform use was common: eight out of 14 users used two or more platforms.

As for the purposes of the use, they were overwhelmingly instrumental and tied to schooling: homework help or studying and information or answers to questions were named by 12 of 14 users, followed by creative projects (6), coding or technical help (5), and ideas for projects or creativity (5). Advice on personal problems, entertainment or casual conversation, and planning or organizing were cited by three users each: in total, just five out of 14 users mentioned any relational purposes of using the chatbot. The open-ended descriptions of the types of exchanges corroborate this description: they involved seeking step by step instructions on mathematics, editing help, quiz generation from class notes, explanation of concepts through lecture slides, prompts to begin the project work, and coding help. One respondent even emphasized the last of them in an instrumental way: "Coding help, because they're made of code after all."

Thus, the usage of the technology in the sample is better characterized as task-oriented and academic rather than companionship-oriented which is an important point when interpreting the expectations data below.

4.3 Relational Expectations Across Conversational Partners

The mean scores for the four relational expectations across five conversational partners are presented in Table 3 and Figure 1. It turned out that there is a hierarchy of human conversational partners' expectations. Family members and close friends attract the highest expectations of all four aspects of the relationship (availability: 4.22 and 3.96; emotional validation: 4.13 and 4.13; conflict tolerance: 3.74 and 3.74; honesty: 4.39 and 4.39), classmates/peers attract the lowest among human partners (3.00, 2.83, 2.96, and 3.43 respectively), and teachers fall somewhere in the middle (3.83, 3.13, 3.65, and 4.22). Against this background, AI occupies an intermediate position, with fourth position among the five partners for availability ($M = 3.74$) and conflict tolerance ($M = 3.30$, above only peers) and last place for emotional validation ($M = 2.13$) and honesty ($M = 3.04$).

There are two noteworthy features of Table 3. Firstly, expectations of AI were more variable than expectations of any human partner: the standard deviation values for the four AI items varied from 1.49 to 1.64, compared to 0.78 to 1.30 for human partners. This means that the respondents were sharply divided about what to expect from AI rather than converging on a similar opinion. Secondly, emotional validation is a dimension where AI differs most from any human partner—its mean value of 2.13 falls 0.70 points below the lowest human partner (peers, 2.83) and nearly 2 points below family and close friends. Section 4.4 shows that these patterns are heavily moderated by use status.

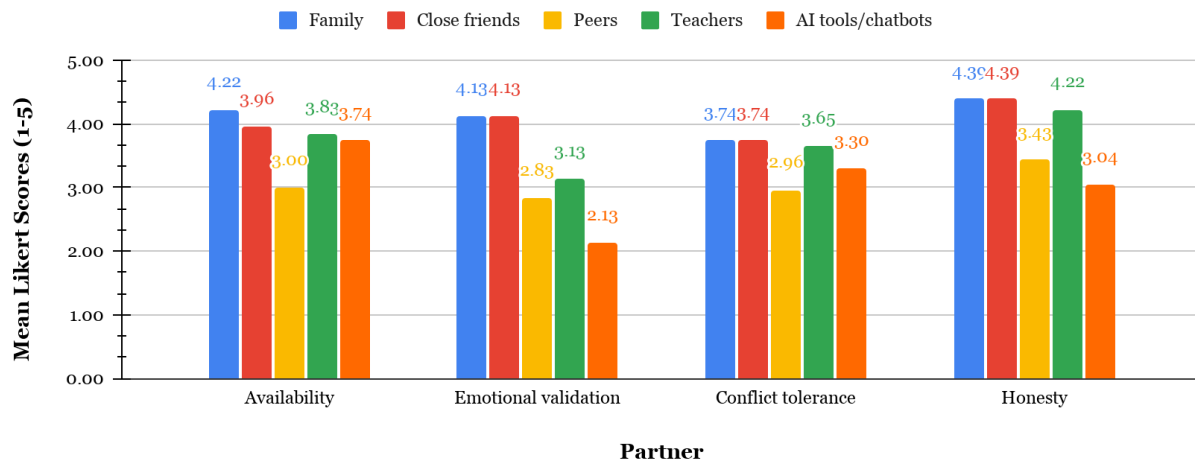
Table 3*Mean Relational-Expectation Scores by Conversational Partner (N = 23)*

Partner	Availability	Emotional validation	Conflict tolerance	Honesty
Family	4.22 (0.80)	4.13 (0.97)	3.74 (0.92)	4.39 (0.78)
Close friends	3.96 (0.88)	4.13 (0.97)	3.74 (0.86)	4.39 (0.78)
Peers	3.00 (0.95)	2.83 (1.03)	2.96 (1.11)	3.43 (0.84)
Teachers	3.83 (1.03)	3.13 (1.22)	3.65 (1.30)	4.22 (0.90)
AI tools/chatbots	3.74 (1.54)	2.13 (1.58)	3.30 (1.64)	3.04 (1.49)

Note: Values are means with standard deviations in parentheses, on 1–5 Likert scales. Higher scores indicate stronger expectations.

Figure 1

Mean Likert Scores For Relational Expectations Across 5 Relational Targets



Note: Mean Likert scores (1–5) for each relational expectation across the five conversational partners, full sample (N = 23).

4.4 Expectations of AI by Use Status: Evidence for H1–H4

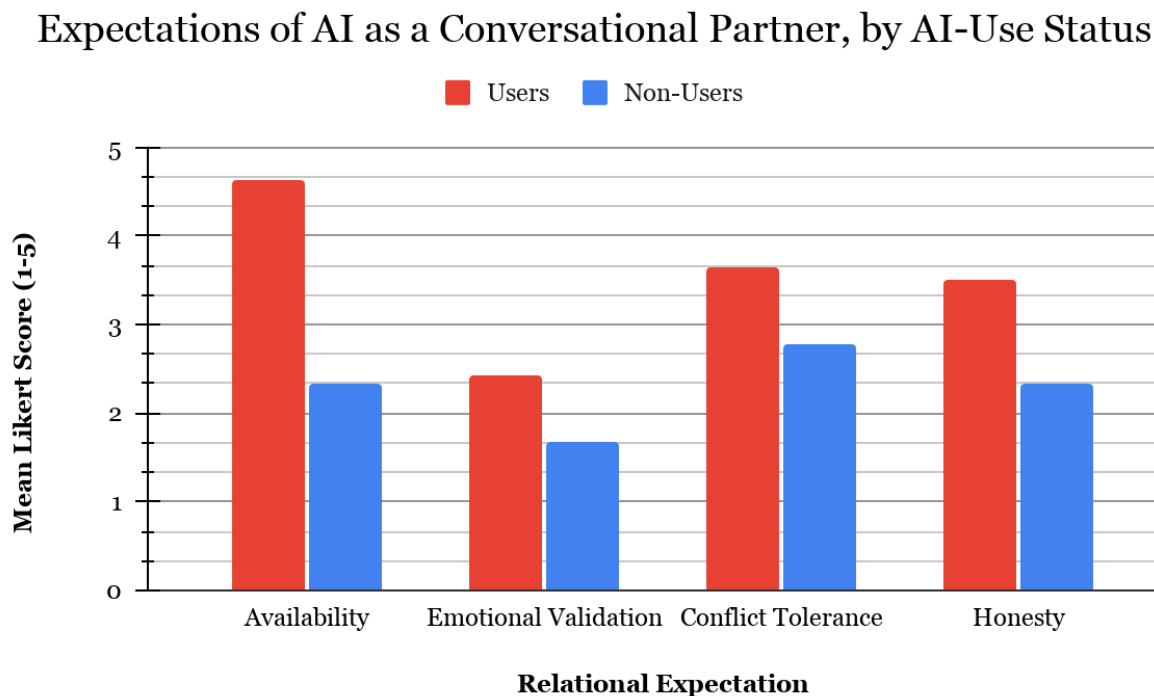
After splitting the sample by use status, most of the aggregate variance in expectations was resolved (Figure 2). The most pronounced difference was seen in availability: users evaluated AI as highly available ($M = 4.64$, $SD = 0.74$), while non-users gave it the lowest score among all five conversational partners ($M = 2.33$, $SD = 1.41$), which is a difference of more than two points on the scale, $t(21) = 5.14$, $p < .001$. Moreover, within the users, AI was evaluated significantly higher in availability than the mean score of the four human partners (4.64 vs. 3.55), paired $t(13) = 3.83$, $p = .002$; it significantly

exceeded the scores for close friends ($p = .021$), teachers ($p = .005$), and peers ($p < .001$), and did not significantly differ from the family ($p = .169$). Hence, H3 holds true for the users—adolescents who use AI expect it to be at least as available as any human relation—but not for non-users, whose expectations of AI availability are lower than every human partner's expectations.

Emotional validation showed a different pattern. AI was the lowest-rated conversational partner in both groups (users: $M = 2.43$; non-users: $M = 1.67$), and even among users it was well below their average for human partners (3.34), paired $t(13) = -2.10$, $p = .055$. Thus, it appears that adolescents grant AI near-unlimited availability while withholding expectations of genuine emotional understanding.

Regarding the expectations of honesty, AI was again the most discounted partner ($M = 3.04$ vs. 4.39 for family and close friends), but while users rated AI as more honest than non-users did (3.50 vs. 2.33), users' rating of AI still was descriptively below their average for human partners (4.02), paired $t(13) = -1.48$, $p = .163$. This relates to H1—the expectations of AI honesty are held low rather than elevated, which likely reflects awareness of hallucinations and sycophantic tendencies. Conflict tolerance showed a similar pattern: users rated AI slightly above their human average (3.64 vs. 3.34) but not significantly so, paired $t(13) = 0.67$, $p = .514$, and nor usage intensity or frequency significantly predict conflict tolerance expectations of AI among users ($r < 0.15$). Despite this, however, usage length seemed to hold a much higher correlation to conflict tolerance expectations in AI ($r = 0.31$). For humans, the case was similar to this as well—usage frequency and intensity had less of a correlation compared to that of length ($r = 0.20$ and $r = 0.14$ vs $r = 0.25$).

Finally, expectations of human partners were examined directly. Users scored descriptively lower than non-users on all four human-target composites, but no comparison reached significance (availability: 3.55 vs. 4.06, $t(21) = -1.97$, $p = .063$; emotional validation: 3.34 vs. 3.89, $p = .110$; conflict tolerance: 3.34 vs. 3.81, $p = .166$; honesty: 4.02 vs. 4.25, $p = .375$), and among users, there were no usage-intensity correlations in overall human expectations ($|r| < .21$). There are no signs in the data that frequent AI use elevates expectations of human partners; if anything, the nonsignificant trend goes in the opposite direction, which is discussed in the next section.

Figure 2

Note: Figure separates respondents who used AI in the past month ($n = 14$) and those who never used AI ($n = 9$).

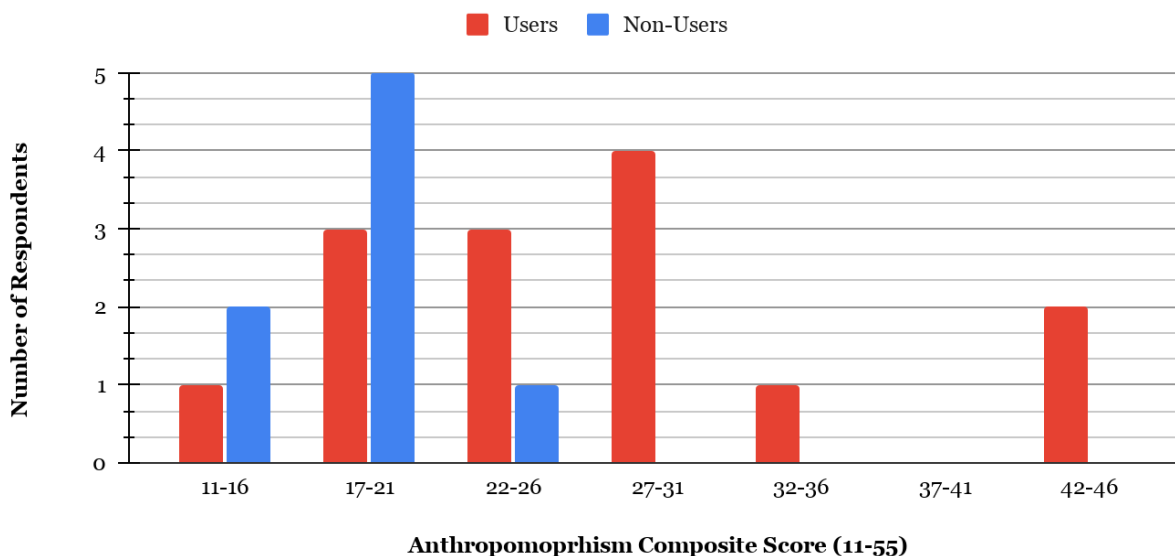
4.5 AIA Scale Performance and the Distribution of Anthropomorphism

In the current sample, the 11-item AIA composite showed good internal reliability (Cronbach's $\alpha = .874$), thus allowing it to be used as the single mediating index. The composite scores ranged quite widely (from 11 to 45 out of a possible range of 11–55) but were mostly distributed around machine perception: the sample mean was 23.0 ($SD = 8.8$, $Mdn = 21$), the interquartile range – from 17 to 27.5, and just two respondents (8.7%), both of them users, scored at or above the scale midpoint of 33 (Figure 3). Thus, on average, the adolescents in this sample described AI in more machinelike terms than humanlike, endorsing statements about real understanding, emotions, and personality very weakly if at all.

Nonetheless, use status moderated the distribution: users scored significantly higher than non-users ($M = 25.9$, $SD = 9.3$ vs. $M = 18.3$, $SD = 5.6$), $t(21) = 2.19$, $p = .040$, and this difference is confirmed by a Mann-Whitney test ($p = .046$). No significant correlations between age or gender and anthropomorphism scores were found, so the demographic covariates specified in the analysis plan were not pursued further.

Figure 3

Distribution of AIA Anthropomorphism Composite Scores, by AI-Use Status



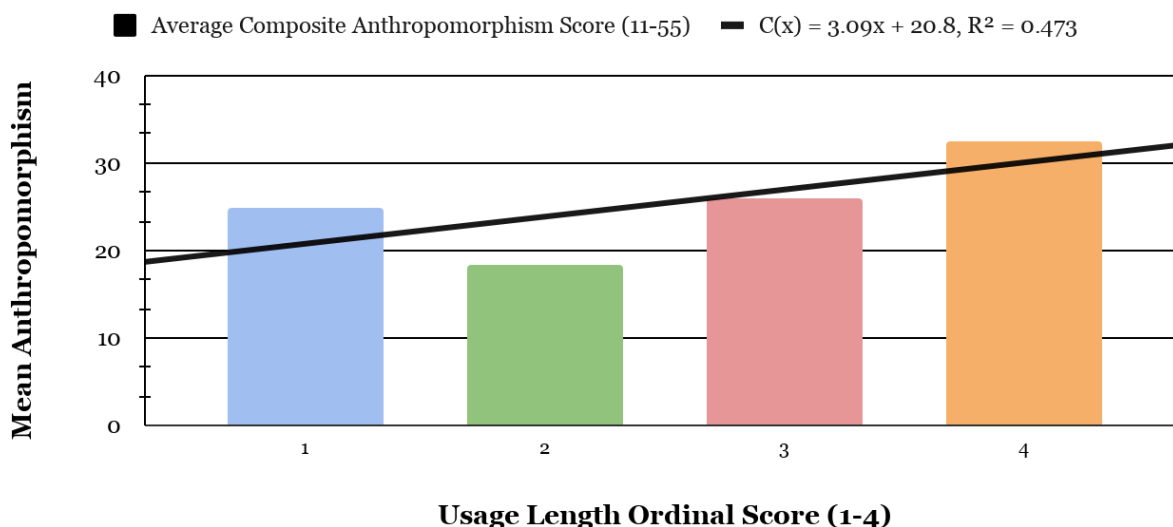
Note: Composite scores range from 11 to 55.

4.6 Usage Intensity and Anthropomorphism

Within the full sample, the usage-intensity index correlates moderately and significantly with the average anthropomorphism composite, $r(21) = .39$, $p = .031$ (Spearman's $\rho = .48$, $p = .021$; Figure 4). This correlation is partially due to the user/non-user split revealed in Section 4.5, thus the two components of intensity were examined separately within the user group ($n = 14$). There, conversation length, but not frequency, proved to be the key component: duration of typical conversation correlated with average anthropomorphism at $r = .69$, while frequency of use was not associated with it ($r = .13$). Thus, adolescents who have longer conversations with AI tend to perceive it as more minded, independent of frequency of use. With such a small sample size ($n = 14$), this trend should be interpreted as suggestive, but it agrees with prior evidence suggesting that deeper engagement with the conversational agent heightens perceptions of person-like properties (Van Brummelen et al., 2021).

Figure 4*Usage Intensity and Anthropomorphism of AI*

Average Composite Anthropomorphism Score (11-55) in Relations to Self-Reported Usage Depth



4.7 Anthropomorphism and Expectations of AI: Testing H5

H5 expected that anthropomorphism would act as a scaling factor on expectations of AI's emotional validation. It is supported by the data. The composite correlates strongly with expected emotional validation from AI, $r(21) = .61$ (Figure 5), and the association holds true among the users alone, $r(12) = .58$, meaning that it is not just a result of the user/non-user split. The two highest anthropomorphizers in the sample (composites of 43 and 45) are also the only two respondents who rated AI's emotional validation at 4 or 5 on the scale.

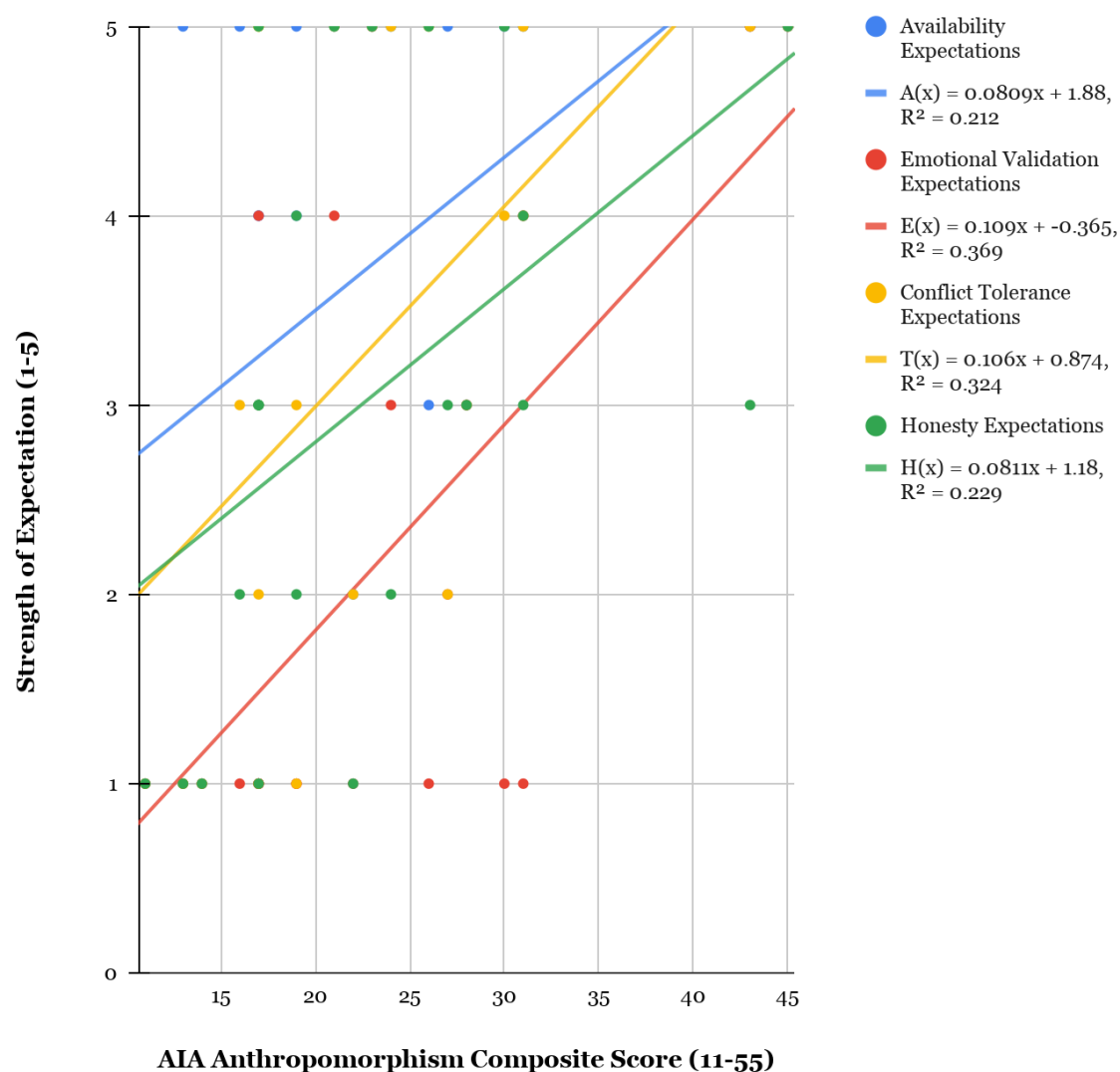
However, the restrictive part of H5, that anthropomorphism matters only for emotional validation, is not supported. In addition to emotional validation, the composite correlates with expected availability ($r = .46, p = .027$), conflict tolerance ($r = .57, p = .005$), and honesty ($r = .48, p = .021$). Anthropomorphism in the current sample behaves not as a valve on a single aspect of expectations but as a general lens: adolescents who see AI as more minded expect more from it in every relational aspect. Nonetheless, the relationship is strongest for emotional validation, and the absolute levels of expectations described in Section 4.4 show that anthropomorphism raises the expectations of AI's validation compared to other users, but nowhere near the levels of expectations for family or close friends. However, noting that the majority of

respondents used AI for task-oriented purposes and viewed AI as machinelike, this revelation could potentially have larger implications for a sample that contrasts this one.

Figure 5

Anthropomorphism and Expected Emotional Validation from AI

Anthropomorphism and Expectatins from AI



Note. Each point is one respondent ($N = 23$); the dashed line is the least-squares fit. The association remains significant among AI users alone, $r(12) = .58$, $p = .030$.

4.8 Retrospective Self-Reported Change (Section E)

Self-report of changes over time told a subtler story than the cross-sectional contrasts. Out of 14 users, 12 reported that their expectations from human relations did not change since they started to use AI regularly, and one selected "I don't use AI tools regularly" option despite the report of the monthly usage in Section B. Just one user reported increased expectations from human relations since he/she started using AI: "Compared to before, I expect more emotional reassurance in conversations." Notably, that respondent was one of the two highest anthropomorphizers in the sample (composite 43) and reported long conversations with AI several times per week—exactly the profile the mediation model predicts to be susceptible to changes. One additional response was internally inconsistent and contained contradictions: it reported no change while endorsing all four change descriptors (expecting faster replies, less patience with slow/unclear communication, more need for emotional reassurance, and making conflicts more exhausting). This response was considered uninterpretable and was excluded from this count.

Therefore, adolescents in the sample do not, in general, perceive any change in their expectations of human relationships. Contrast between this overwhelming subjective stability and the significant AI vs. human benchmarking observed among users in Section 4.4 suggests that whatever real change takes place happens below the level of conscious awareness – a pattern that has immediate implications for the design of digital literacy interventions. All nine non-users selected either "I don't use AI tools regularly" or "My expectations haven't changed."

4.9 Qualitative Themes

Uptake of the optional open-ended items was sparse, thus the qualitative section is used here mostly for triangulation, but still some themes can be found clearly.

Instrumental, school-anchored engagement

Free-text descriptions of the most frequent exchanges by the users focused almost exclusively on school-related activities and information: "to show me how to answer math questions with step by step instructions," "help me edit my things," generating "quizzes from the class notes," "understanding tricky concepts," and "prompts to help me get started on projects." Even when describing the sustained engagement (debates, idea generation), the framing was instrumental. This agrees with the checkbox data in Section 4.2 and explains the absence of companionship-type effects in this sample: most of these adolescents do not use AI as a companion, a possible product of the school setting the survey was distributed in.

Machinelike framing among non-users

All three substantive comments from the non-users were remarkably homogeneous in denying AI any emotions or moral interior: "AI cannot feel empathy or any real human emotion," "AI can't care," and "AI does not have emotions. The only emotion I have for AI is hatred." These statements agree with the quantitative findings regarding lower anthropomorphization of AI (Section 4.5) and lower expectations from it (Section 4.4) among non-users, and suggest that in this sample non-use is somewhat principled rather than mere lack of use.

Perceived stability of human expectations

Consistently with Section 4.8, no respondent used the open-ended change item to describe a transformation in their human relations. The single affirmative retrospective account – increased expectations of "more emotional reassurance in conversations" – came from a high-anthropomorphizer with long conversations with AI, thus qualitatively agreeing with the logic of the mediating hypothesis H5.

SECTION 5: DISCUSSION & CONCLUSION

5.1 Evaluation of Hypotheses

This research sought to determine whether usage intensity (defined via ordinal measures of conversation length and frequency) predicted adolescents' relational expectations with regard to availability, emotional validation, conflict tolerance, and honesty, and whether anthropomorphism mediated that relationship. The findings partially support the hypotheses presented in the theoretical background. They indicate that adolescents who interact with AI have a unique profile in expectation toward it, one characterized by nearly maximal availability, an average level of conflict tolerance, lesser levels of honesty, and an extremely restricted ability to emotionally validate. Such expectations are highly distinguishable from those held by adolescents who refrain from interaction with AI in every aspect examined above.

The strength of anthropomorphization toward AI positively predicts adolescent expectations towards their AI across each of the four aspects mentioned above. Importantly, the relationship is strongest when predicting emotional validation. The lack of the proposed spill-over effect (except in either cross-sectional comparison or self-report assessment) cannot serve as evidence for rejection of the theoretical framework proposed in the discussion since this particular sample of respondents utilize AI for instrumental reasons concerning academic activities, thus portraying

machine-like characteristics when describing AI. This section will assess H1-H5 (Table 4), address the research question, and provide study limitations and directions for future studies.

Table 4

Evaluation of Study Hypotheses

Hypothes is	Prediction	Key evidence	Verdict
H1: Honesty	AI honesty ratings considerably lower than human counterparts and does not change from use	AI rated least honest of all partners ($M = 3.04$); users' AI-honesty rating (3.50)* is below their human average (4.02)*, but differs from non-users (2.33)*	Partially Supported
H2: Conflict tolerance	Conflict tolerance significantly correlated with frequency of use	Weak association between usage and conflict-tolerance expectations of AI ($r = 0-0.31$)* or of humans ($r = 0.14-0.25$)*	Not supported/Weakly supported
H3: Availability	AI deemed equally or more available than all human relations	Among users, AI availability (4.64) exceeded the human-partner average, $t(13) = 3.83$, $p = .002$; among non-users it ranked last (2.33)	Supported among users
H4: Emotional validation	AI emotional validation rated low	AI lowest-rated partner overall ($M = 2.13$) and within both user and non-user groups	Supported
H5: Anthropomorphism	Anthropomorphism scales emotional-validation expectations only	Strongest correlation with validation ($r(21) = .61$), but availability, conflict tolerance, and honesty also correlated ($r = .46-.57$)	Partially supported

Note: All correlations are Pearson coefficients for the full sample ($N = 23$) unless otherwise noted by “*”, which means the statistic pertains to a group in the sample.

Interpretation of H1

H1 hypothesized that AI honesty would be rated low compared to human counterparts regardless of use status, and that AI use would have no impact on honesty expectations. Section 4.2, and 4.9 highlight how both the quantitative and qualitative responses corroborate in the result that AI is more or less viewed as a tool rather than a companion amongst respondents. Honesty measurements add intricacy by depicting that users and non-user alike both view AI as a tool with the capacity of making mistakes. Park et al. (2024)—who mentions AI deception—and Beach (2025)—who examines AI's struggle in discerning fact from belief—both document AI's limitations.

Adolescents' low honesty rating (which supports the first half of H1) could be an adaptation to similar documented limitations.

The part of the hypothesis that was not supported was that usage status would not have an effect on AI honesty expectations. The expectations were higher among users (3.50) but still considerably lower than the expectations they had from all other targets except for peers/classmates (3.21).

Why AI is rated higher for honesty in the user pool might arise from two explanations:

1. Some users who have seen limitations of AI's honesty could've pulled away, subsequently being in the non-user group. Though this seems plausible at first, it is of small likelihood, given that a survey of 1000+ users of even as simple an AI client as Google AI reported that "just 7.71% report always following the links provided by AI Overviews, and only 10.97% usually do so" (Martin, 2026).
2. How factual AI is deemed by users might carry over from how much they anthropomorphise the chatbot (Kara, 2025).

Interpretation of H2

H2, which hypothesizes that conflict tolerance expectations are significantly correlated to usage frequency, was not supported. One important distinction should be made about the complexity sycophancy adds to rating conflict tolerance: AIs have a high tendency to mirror the beliefs of the user (Jain et al., 2026), which respondents could reasonably interpret as the highest amount of conflict tolerance possible or the absence of conflict as a whole. What reinforces this claim is that AI conflict tolerance carried the highest standard deviation of any AI item in Table 3 ($SD = 1.64$). The construct still exhibits correlation, though, seen by how anthropomorphism had a notable correlation with conflict tolerance expectations ($r = .57, p = .005$). The high standard deviation and the presence of strong correlations regardless, in turn, blurs the extent to which the survey item may have been interpreted in multiple ways.

Interpretation of H3

H3 expects that adolescents will view AI as equally or more available than all human counterparts. There was a large discrepancy between users and non-users in this regard: Users rated the expectations they had for AI availability higher than any human counterpart ($M = 4.64$), while non-users rated AI as the least available out of any counterpart ($M = 2.33$). This user/non-user split for availability might originate from the lack of prevalence AI has in non-users' lives—since non-users do not use AI, it poses no available position within their own relational landscapes. Considering this

perspective provides insight into how respondents may have approached the survey and why the availability scores for non-users were so low relative to other counterparts while the standard deviation was almost double that of users scores ($SD = 1.41$ vs $SD = 0.74$).

Interpretation of H4

It is hypothesized in H4 that AI would emerge as the lowest rated for emotional validation compared to any counterpart. Without doubt, the results provided strong support for this hypothesis. As seen in Table 3 and illustrated in Figure 2, AI was rated at $M = 2.13$ regarding emotional validation, scoring 0.70 points lower than peers ($M = 2.83$) and nearly two whole points lower than family members and close friends ($M = 4.13$). Such an availability-validation dissociation—where adolescents view AI as constantly available but emotionally lacking—refers to the existence of an implicit boundary placed by adolescents when attributing machine "understanding". Indeed, this outcome aligns well with the results reported by Kim et al. (2018), who found teenagers appreciated conversational agents precisely because of their lack of emotional interiority. Similarly, qualitative responses gathered in the current study reinforce the boundary phenomenon—non-users explicitly deny any ability of AI to understand emotions, whereas even users frame the interaction with the technology instrumentally rather than emotionally. This is reinforced by the qualitative response of two non-users, who said "AI cannot feel empathy or any real human emotion" and "AI does not have emotions" respectively. Consistently low emotional validation ratings across user and non-user groups indicate, at least within the current sample, that boundary erosion observed by Jacobs et al. (2024) does not extend itself to emotions. Adolescents appear ready to delegate their availability and informational needs to AI but preserve humans exclusively for the latter.

Interpretation of H5

H5 postulated that anthropomorphism would play a role of a scaling factor with regard to the expectation of emotional validation specifically. Though the correlation is confirmed by the results presented in the discussion—there being a strong relationship between anthropomorphism and emotional validation ($r(21) = .61$, $p < .001$; Figure 5) – the narrower prediction concerning exclusive mediation on the former variable was not supported. Instead, anthropomorphism positively predicts AI expectations regarding availability ($r = .46$, $p = .027$), conflict tolerance ($r = .57$, $p = .005$), and honesty ($r = .48$, $p = .021$). Taken together, these associations suggest a more holistic pattern: adolescents who view AI in terms of mental capacities have higher relational expectations from the agent overall. Such findings echo the results obtained by Colombatto et al. (2025), which demonstrated distinct downstream outcomes arising from specific mental state attributions. In the sample of this study, the visualized

correlation, which applies to all relational expectations, implies that high anthropomorphizers have a much higher likelihood of blurring the line between tool and companion. Nonetheless, even the highest of anthropomorphizers in this sample rated the emotional validation of AI as lower than that of family members and close friends, therefore suggesting that AI anthropomorphism increases expectations of AI relative to other users without undermining the fundamental human-AI split.

5.2 Answering the Research Question

Combining these insights helps formulate the response to the primary research question raised in the study. On the one hand, usage intensity, particularly conversation length, predicts anthropomorphism and heightened expectations towards AI in general. On the other hand, the anticipated spill-over effect—that AI interactions recalibrate adolescents' expectations toward human partners—did not manifest itself in this particular sample. No significant differences emerged in cross-sectional comparisons of user versus non-user expectations toward human partners, and retrospective self-reports indicated consistent stability (Section 4.8). Given such results, lack of spill-over may result from a sample bias favoring instrumental over companion-oriented usage (Section 4.2). When used purely to accomplish school-related tasks, AI is compartmentalized into being a tool, separating its relational role from humans. Nevertheless, a lone participant high in anthropomorphism reported higher needs for reassurance in human conversations (Section 4.8), implying that the mediational model formulated in this study could operate under specific circumstances (namely, when AI is used as a companion).

5.3 Study Limitations

Many limitations are worth mentioning for the scope of this study. First is the small sample size ($N = 23$) drawn from only one Canadian high school, which may make this study incapable of detecting small yet significant spill-over effects onto human relational expectations, something that would require a considerably larger sample size and/or a more powerful statistical technique. Descriptively, users tended to report somewhat lower scores in comparison to non-users for various human-target relational expectations—though not reaching significance levels—requiring replication in subsequent studies. Next up, the cross-sectional design employed restricts conclusions about causation: just because those who had longer conversations were more anthropomorphic in nature didn't mean that those who increase their conversation length will become more anthropomorphic. Longitudinal designs would help address such issues adequately. Moreover, the prevalent use of AI as a tool amongst respondents renders this particular study unable to capture the full extent of companion-related expectations, a potential major predictor of spill-over effects. Furthermore, subjective

self-reports concerning retrospective change in expectations (almost all of which were subjectively stable) likely suffer from bias via poor awareness of subtle shifts in expectations. Finally, although showing high reliability ($\alpha = .874$), the new AIA scale requires additional psychometric validation—ideally, against well-established scales of mental-state perception.

5.4 Future Work & Contributions

Possibilities for Future Research

These limitations suggest several rewarding research directions moving forward. Longitudinal experimental designs resembling those implemented by Guingrich and Graziano (2025), adapted to adolescents, would prove *Possibilities for Future Research* immensely helpful in tracing actual changes in relational expectations caused incrementally by AI companion use. Experiments manipulating conversation length and frequency separately could clarify whether this study's findings concerning the impact of length over frequency for anthropomorphism correlations generalize beyond cross-sectional design, therefore addressing the question of whether AI anthropomorphism is an innate human property or a measure that can be distorted with different usage patterns. More broadly, future studies should try specifically sampling companion users, better helping assess spill-over effects. Another point of future research is to measure the AIA scale's robustness against demographics differing from the Canadian one in the study. Finally, the theoretical contribution of Colombatto et al. (2025)'s distinction between experiential and cognitive anthropomorphism should be applied to adolescent samples: do teens who attribute emotional qualities to AI experience different relational outcomes than teens who solely attribute intelligence?

Summary of Contributions to the Field

Despite the limitations, the present study makes several substantive contributions to the emerging literature on adolescent AI use. First of all, introducing Artificial Intelligence Anthropomorphism (AIA) scale and establishing its psychometric qualities (specifically reliability $\alpha = .874$) provides a crucial stepping stone for future empirical endeavors. By building upon the Robot's Perceived Empathy (RoPE) scale by extending it to four dimensions common in AI interactions, the novel AIA measure offers an important tool previously unavailable for studying AI companion perception. Secondly, by providing data that anthropomorphism affects all four relational dimensions assessed in the study, it manages to complicate H5. Thirdly is that the study manages to build off of Van Brummelen et al. (2021)'s by suggesting that usage length matters more than habitual exposures for mind perception in AI. The fourth contribution involves documenting a reliable availability-validation dissociation (the fact that AI was largely expected to be

available but not emotionally capable), a neat detail that challenges the perspective that anthropomorphism is applied uniformly. Finally, the interpretation of the tool/companion distinction, the idea that instrumental AI use dampens spill-over effects of use, presents a testable idea for future studies to address.

As AI use becomes more normalized in the adolescent demographic, this study presents an optimistic idea of how it could affect one's other relationships. The findings suggest that adolescents have not allowed their AI use, which was largely instrumental, to affect their human relationships considerably. Nonetheless, what makes up the lack of spill-over effects—instrumental use, low baseline anthropomorphism, etc.—is exactly what could be at risk as AI is embedded into daily life. Whether the boundary of machinelike views of AI upholds for future generations is now the big question.

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APPENDICES

All materials used in the formation and dissemination of the survey are in the folder below:

<https://drive.google.com/drive/u/o/folders/1QPUKFL7ovkHeLHbXFmYhd7d9UhSDQmrr>